Ethology **104**, 695--708 (1998)
© 1998 Blackwell Wissenschafts-Verlag, Berlin
ISSN 0179-1613

School of Biological Sciences, Nicholson Building, University of Liverpool and Department of Anthropology, University College London, UK

Neocortex Size Predicts Group Size in Carnivores and Some Insectivores

R. I. M. Dunbar & Julian Bever

Dunbar, R. I. M. & Bever, J. 1998: Neocortex size predicts group size in carnivores and some insectivores. Ethology **104**, 695–708.

Abstract

Neocortex size has been shown to correlate with group size in primates. Data for carnivores and insectivores are used to test the generality of this relationship. The data suggest that carnivores lie on the same grade as the primates, but that insectivores lie on a separate grade to the left of these two orders. Among the insectivores, there appears to be a distinction between the 'advanced' genera (which show a relationship between group size and neocortex size) and the 'basal' genera (which do not).

Corresponding author: R. Dunbar, School of Biological Sciences, Nicholson Building, University of Liverpool, Liverpool L69 3BX, UK

Introduction

Several recent studies have demonstrated a relationship between the relative size of the neocortex and group size in primates (Sawaguchi & Kudo 1990; Dunbar 1992, 1995; Barton 1996). The robustness of this relationship raises the question of whether it is unique to primates. Primate social groups are recognized as being structured in rather different ways from those of other species: they are based on intense social bonds and the formation of long-term coalitions, both of which are founded on deep social knowledge of the past behaviour and, perhaps, strategic interests of other individuals (including third parties). Primate groups appear to be distinguished by the extent to which 'mind-reading' is an essential component of the cognitive processes involved in their make-up (Byrne & Whiten 1988; Harcourt & de Waal 1992).

If primate groups are, in this sense, unique then we would not expect the relationship between neocortex size and group size to be found in non-primate species. Alternatively, it may be that this relationship holds true for all (or most) mammals because the underlying cognitive processes involved in social life are essentially the same in all species, even though primates might take them to a more extreme degree. Some evidence to suggest that the second possibility might be the case is offered by the fact that there appear to be distinct grades in the neo-

U. S. Copyright Clearance Center Code Statement: 0179-1613/98/1048–0695$14.00/0

cortex/group size relationship even within primates: prosimians appear to be able to support larger groups for a given volume of neocortex than monkeys, who in turn are able to support larger groups than apes (Dunbar 1993, 1998). Since this sequence mirrors rather closely the perceived levels of complexity in primate social life, it seems plausible to suppose that other mammals may fit the same pattern. Preliminary analyses for cetaceans suggest that this relationship might indeed be general (Marino 1996; Tschudin 1996). However, lack of adequate data forced both of these studies to use estimates of group size in cetaceans that are closer in definition to the structurally loose herds characteristic of ungulates rather than the socially cohesive groups of the primates.

In this paper, we ask whether the hypothesized relationship between neocortex size and group size applies to two other mammalian orders, the Carnivora and Insectivora. The carnivores represent a relatively recently evolved rather diverse group of species, many of which exhibit social skills comparable to those of at least some primates (e.g. the social carnivores that hunt in groups). They therefore seem to be an obvious group with which to assess the generality of the primate relationship. Some evidence in support of the hypothesis is provided by Gittleman's (1986) finding that relative brain size is related to breeding system in carnivores. However, the social brain hypothesis focuses on neocortex size rather than total brain size (Joffe & Dunbar 1997), and in this paper we therefore seek to determine whether neocortex size predicts group size.

In contrast, the insectivores provide a baseline against which to consider the magnitude of mammalian evolution (Stephan 1972; Stephan et al. 1981). Although the insectivores are something of a taxonomic dustbin into which any small mammal whose taxonomic affinities are not immediately clear is consigned, it is generally accepted that the insectivores, as a whole, represent the least evolutionarily derived group of species and are thus the taxon that is likely to be most representative of the group of primitive mammals from which all modern mammals evolved. One drawback with respect to using the insectivores in this kind of analysis is that, being both nocturnal and often fossorial, their behaviour (in particular, social group size) is often poorly known.

Methods

Data on carnivore brain size were obtained from two sources. Those for Mustelidae are based on Röhrs (1986a), while those for the Ursidae and the Procyonidae are based on Kamiya & Pirlot (1988a,b) and Röhrs (pers. comm.). Data on group size are based on Gittleman (1989a,b). These data are given in Table 1 for those species for which both sets of data are available. Unfortunately, only a few of the species in Table 1 are social and this makes detailed testing of the hypothesis difficult. However, Aiello & Dunbar (1993) found that, in primates, relative neocortex size can be predicted quite reliably from total brain size. If a comparable equation can be derived for carnivores, we might be able to increase the sample of social species.

Table 1: Neocortex size and group size for carnivores for species with known neocortex volume. Data from Röhrs (1986a and pers. comm.) and Kamiya & Pirlot (1988a,b)

Species	Brain volume (mm^3)	Neocortex volume (mm^3)	C_R[1]	N[2]
Ailuropoda melanoleuca	211809.3	136435.7	2.355	1
Ailurus fulgens	48735.88	28692.8	1.414	2
Conepatus chinga	17822	9228	1.074	1
Conepatus humboldti	11682	5735.9	0.965	1
Eira barbara	45688	29077	1.750	4
Galictis cuja	17944	10726	1.486	2
Helarctos malayanus	345250	252159	2.709	1
Martes foina	20857	11594	1.252	1
Meles meles	56771	32604	1.349	7
Mustela erminea	4443	2101.1	0.897	1
Mustela nivalis	1795	762.16	0.738	1
Mustela putorius	8899.6	4147	0.873	1
Procyon lotor[3]	41657.6	24735.51	1.407	1
Pteronura brasiliensis	115203.1	83257.3	2.606	5.5
Ursus americanus	140347.5	88420.25	2.439	1

[1] Ratio of neocortex volume to volume of rest of brain (i.e. total brain volume minus neocortex volume).
[2] Mean group size (Gittleman 1989a,b).
[3] From M. Röhrs (pers. comm.).

The reduced major axis regression equation for carnivores yielded by the data in Table 1 is:

$$\log_{10}(N_V) = 1.1024 \log_{10}(B_V) - 0.7059 \quad (1)$$

($r^2 = 0.999$, $n = 18$, $p < 0.0001$), where N_V is the neocortex volume and B_V is the total volume of the brain (both in mm^3). We have therefore used this equation in conjunction with Röhrs's (1986b) database on brain volume for mustelids and that for canids and felids from Röhrs et al. (1989). These data are given in Table 2. It is clear from comparisons of the different databases, e.g. Gittleman (1986) vs. Röhrs et al. (1989b), that different methods of measuring brain volume can produce significant differences in estimated brain volume. In order to avoid difficulties involved in scaling data from different sources, we have limited our sample to the species for which brain volume data are given by Röhrs et al. rather than using larger databases available from other sources because these authors give actual brain volumes (as opposed to intra-cranial capacities). In order to maintain comparability between data sets, we also recalculated neocortex volumes for those species given in Table 1 using equation (1).

The insectivore dataset was taken from Frahm et al. (1982) and Stephan et al. (1981, 1984). These sources provide data on brain volume, as well as on the volumes of other brain parts (in particular the neocortex and the telencephalon). Table 3

Table 2: Estimated neocortex size and group size for carnivores. Data from Röhrs (1986b) and Röhrs et al. (1989)

Species	Brain volume (mm^3)	Neocortex volume[1] (mm^3)	C_R[2]	N[3]
Alopex lagopus	31742	17769	1.27	2
Canis adustus	47416	27640	1.40	2
Canis aureus	60708	36281	1.49	3.3
Canis latrans	85976	53217	1.62	2.1
Canis lupus	131519	84973	1.83	7
Canis mesomelas	53109	31314	1.44	2
Chrysocyon brachyurus	113391	72173	1.75	1
Conepatus chinga	17843	9404	1.12	1
Conepatus humboldti	11675	5909	1.03	1
Crocuta crocuta	160919	106105	1.94	55
Cuon alpinus	93900	58641	1.66	6.3
Eira barbara	45662	26517	1.39	4
Galictis cuja	17919	9469	1.13	2
Galictis vittatus	17347	9137	1.11	2
Helictis personata[4]	14296	7384	1.07	1
Hyaena brunnea	122040	78256	1.79	9
Hyaena hyaena	84889	52477	1.62	1
Lutra canadensis	56799	33718	1.46	3.2
Lutra lutra	44956	26066	1.38	2.8
Lycaon pictus	128381	82744	1.81	8
Martes foina	20760	11133	1.16	1
Martes martes	16727	8778	1.10	1.5
Martes pennanti	32686	18352	1.28	1
Meles meles	56779	33705	1.46	7
Mephitis mephitis	10998	5532	1.01	1
Mustela erminea	4450	2043	0.85	1
Mustela nigripes	7433	3594	0.94	1
Mustela nivalis	1810	759	0.72	1
Mustela putorius	8796	4326	0.97	1
Mustela vison	9140	4512	0.98	1
Nasua rufa	37625	21427	1.32	8
Nyctereutes procyonoides	28863	16004	1.24	2
Otocyon megalotis	26909	14815	1.23	2
Panthera leo	215021	145983	2.11	8.7
Potos flavus	29340	16295	1.25	2
Procyon lotor[5]	40943	23516	1.35	1
Proteles cristatus	34650	19570	1.30	1
Pteronura brasiliensis	116080	74060	1.76	5.5
Taxidea taxus	59878	35736	1.48	1
Urocyon cineroargenteus	35556	20134	1.22	2
Vulpes rüppelli	26689	14682	1.27	3.8
Vulpes velox	31742	17769	1.31	1.5
Vulpes vulpes	48332	28229	1.40	3

[1] Estimated from total brain volume using equation (1).
[2] Ratio of neocortex volume to volume of rest of brain (i.e. total brain volume minus neocortex volume).
[3] Mean group size (from Gittleman 1989a,b).
[4] Not available for analysis by independent contrasts.
[5] From M. Röhrs (pers. comm.).

Table 3: Neocortex size and group size of insectivores. Data from Frahm et al. (1982) and Stephan et al. (1981, 1984)

Genus	Cerebral development[1]	No. of spp. sampled	Brain volume (mm^3)	Neocortex volume (mm^3)	C_R[2]	N[3]
Atelerix	Unknown	1	2859	500	0.212	1
Blarina	Unknown	1	378	69.7	0.226	1
Chlorotalpa	Advanced	1	693	129	0.229	1
Chrysochloris	Unknown	1	657	113	0.208	1
Crocidura	Basal	6	269	36.0	0.154	1
Desmana	advanced	1	3620	983	0.373	4
Echinops	Basal	1	566	51.5	0.100	1
Elephantulus	Advanced	1	1233	236	0.237	2
Erinaceus	Basal	1	3050	522	0.206	1
Galemys	Advanced	1	1230	339	0.380	2
Hemicentetes	Basal	1	757	73.4	0.107	2
Hemiechinus	Unknown	1	1710	309	0.221	1
Microgale	Unknown	3	548.7	72.1	0.153	1.17
Micropotamogale	Unknown	1	743	144	0.240	1
Microsorex	Unknown	1	96.9	16.2	0.201	1
Neomys	Advanced	1	295	54.2	0.225	1
Oryzorictes	Unknown	1	538	66.5	0.141	1
Potamogale	Advanced	1	3878	1054	0.373	2
Rhynchocyon	Advanced	1	5680	1089	0.237	3.4
Setifer	Basal	1	1404	134	0.106	1
Solenodon	Advanced	1	4282	661	0.183	1
Sorex	Basal	4	174.3	29.3	0.195	1
Suncus	Basal	2	205.3	28.3	0.151	1
Talpa	Advanced	1	953	181	0.234	1
Tenrec	Basal	1	2315	271	0.133	1

[1] After Stephan (1972).
[2] Ratio of neocortex volume to volume of rest of brain (i.e. total brain volume minus neocortex volume), averaged for all species sampled in genus.
[3] Mean group size (source: general insectivore literature).

gives the data for all insectivore genera for which data on both brain part volumes and social group size are available. Although Stephan et al. (1991) provide a much more extensive database on insectivore brain part volumes, no data are available on social group size for any of the species included in this database that are not already included in the two earlier data sets.

Most insectivore species are nocturnal and forage solitarily; consequently, group size data are likely to be especially subject to census biases in the field (poor censusing opportunities in difficult conditions, failure to keep neighbouring groups separate in areas of range overlap, etc.). In view of this, we did not include data

suggesting large groupings when these were based on a single study. In addition, offspring were discounted when calculating mean group size unless it was explicitly stated that offspring remained after the birth of the next litter.

We have followed Dunbar (1992, 1998) in using the neocortex ratio (C_R, defined as neocortex volume divided by the volume of the rest of the brain) as the best measure of relative neocortex size. Indices that use body mass as a baseline for comparison risk being confounded by the fact that, during evolution, body size frequently changes without any corresponding change in brain size (Willner & Martin 1985; Willner 1989; Deacon 1990). As a result, it is often difficult to know whether changes in relative brain size result from changes in absolute brain size or changes in body size. It appears likely that, as suggested by Jerison (1973), there is a relationship between body size and non-cortical brain volume (reflecting the quantity of muscle mass and other life-supporting functions that have to be managed); cortical (or neocortical) volume may then respond to sensory or social computational demands in a way that is much less dependent on body size (Joffe & Dunbar 1997). It is therefore important not to confound changes in brain size with changes in body size (Barton & Dunbar 1996). We note that, since all data are logged, the neocortex ratio is identical to the residuals from total brain volume commonly used in many analyses of this kind (see Barton 1996), differing only in the base against which the neocortex is relativized (Joffe & Dunbar 1998). In the case of neocortex ratio, we use the observed (or estimated) volume of the non-cortical brain, whereas conventional residuals methods (e.g. Barton 1996) use a smoothed estimate based on the overall pattern for the whole taxon.

Since data from many closely related species are compared in these analyses, we used approved phylogenetic methods for analysing the data wherever possible. For this purpose, we used the method of independent contrasts recommended by Harvey & Pagel (1991), in which contrasts in group size are regressed against contrasts in neocortex ratio. This method makes pairwise comparisons between the differences in each of two variables (in the present case, neocortex ratio and group size) at various levels in the phylogeny. Values for nodes above the level of individual species are obtained by averaging the values for the immediately lower pair of nodes. We made no allowance for branch lengths in the phylogeny since these are generally too poorly known. Note that the phylogenies available to us are incomplete, so that only certain of the species listed in Tables 2 and 3 can be used in independent contrasts analyses.

We used the phylogeny given by Bryant et al. (1993) for the mustelids, with the phylogeny obtained by Zyll de Jong (1987) for the Lutrinae inserted into it, and that by Wayne et al. (1987) for canids, felids and other taxa. Whenever there was disagreement between the genetic and morphological evidence concerning the taxonomic affinities of different groups (e.g. that between the ursids, including the giant panda and the red panda, *Ailurus*: compare Wayne et al. (1989) with Wozencraft (1989)), we gave priority to the genetic data. All these phylogenies still yield a number of unresolvable cases where three or more species branch from a given node; in these cases, we carried out pairwise comparisons of successive species in the order given on the phylogenetic tree (i.e., A with B and B with C, but not A

with C). This avoids wasting too much data without compromising statistical independence. The mean value for the higher node in these cases is the average for all the species branching from that node. Insectivore taxonomy is generally considered to be unsatisfactory and no widely accepted phylogenies are available. We are therefore unable to carry out independent contrasts analyses for this taxon.

Stephan (1972) distinguishes 'basal' from 'advanced' insectivores in terms of brain size and structure. The 'basal' insectivores include such groups as tenrecs, hedgehogs and shrews, while the 'advanced' group includes elephant shrews, desmans, water shrews and related species. Stephan (1972) classified only those species for which he had brains to analyse, so the classification of insectivores as 'basal' or 'advanced' given in Table 3 extends only to those species included in his sample.

In order to compare the results directly with those of Dunbar (1992), we also undertook conventional bivariate analyses of generic means. Since genera commonly represent different *Baupläne* at the ecological and/or reproductive levels, the use of generic means goes some way to obviating the problems associated with phylogenetic inertia (albeit at some cost in terms of reduced sample size). One of our concerns is to determine whether or not insectivores and carnivores are similar to primates, and it is easier to detect grade shift differences (that is, differences in intercept but not slope) with a generic means analysis than with an independent contrasts analysis.

Whenever regression analyses were undertaken to obtain functional equations, we determined the reduced major axis (RMA) equation. The RMA is considered to be the most appropriate choice in those cases (such as the present) where there is uncontrolled error variance on both variables (Rayner 1985). Tests of independent contrasts analyses, however, were carried out using linear regression set through the origin, as recommended by Harvey & Pagel (1991).

In all cases, a specific hypothesis (that group size correlates positively with neocortex ratio) is being tested, and one-tailed statistical tests are therefore appropriate. All p-values are therefore given as one-tailed, unless stated otherwise. A negative correlation between the two variables has the same logical status as zero correlation.

Results

Fig. 1 plots generic means for group size against neocortex ratio for those carnivore genera for which known brain part volumes are available (Table 1 dataset). These consist mainly of mustelids, with a small number of ursids, procyonids and pandas. The best-fit RMA regression equation is:

$$\log_{10}(N) = 2.102 \log_{10}(C_R) - 0.658$$

($r^2 = 0.288$, $t_{13} = 2.29$, $p < 0.05$), where N is the mean group size and C_R is the neocortex ratio (neocortex volume divided by the volume of the rest of the brain). The weakness of the relationship is due mainly to the large group size value for

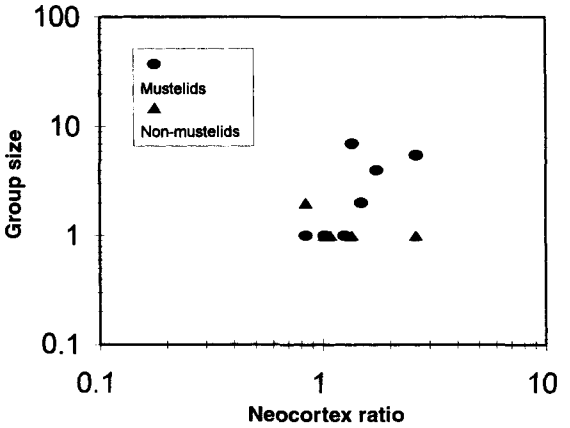


Fig. 1: Mean group size plotted against actual neocortex ratio for carnivore genera (see Table 1)

Meles meles relative to its neocortex size and the large neocortex ratios relative to group size characteristic of the ursids and the giant panda.

Fig. 2 plots independent contrasts from the same data set. There are 14 contrasts available from the 15 species for which a phylogeny can be specified (with two unresolvable nodes of three taxa each). The least squares regression equation through the origin is:

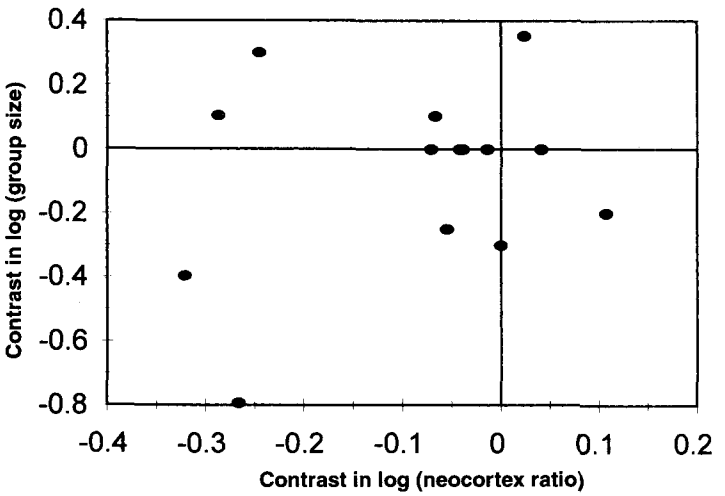


Fig. 2: Independent contrasts in group size plotted against contrasts in actual neocortex ratio for the mustelid species given in Table 1, using the phylogeny of Bryant et al. (1993) and Zyll de Yong (1987)

Contrast in $\log_{10}(N) = 0.660$ contrast in $\log_{10}(C_R)$

($r^2 = 0.124$, $t_{12} = 1.356$, $p = 0.099$). *M. meles* and the ursids/pandas are again the main sources of discrepancy.

Fig. 3 plots generic means for group size against neocortex ratio for all the carnivore genera in Röhrs's sample of brain volumes for which neocortex volumes can be estimated using equation (1) (see Table 2). The best-fit RMA regression equation is:

$$\log_{10}(N) = 5.772 \log_{10}(C_R) - 0.459$$

($r^2 = 0.400$, $t_{14} = 4.792$, $p < 0.001$).

It is possible to carry out an independent contrasts analysis to check for phylogenetic effects using 40 of the carnivore species listed in Table 2 by combining the phylogeny given by Bryant et al. (1993) for mustelids with that given by Wayne et al. (1989) for canids and felids. (Note that neocortex ratios for all the species concerned, including the mustelids and ursids in this sample, are based on estimated neocortex volume: in contrast, the independent contrasts analysis shown in Fig. 2 is based on actual neocortex volume. Note also that in order to maintain consistency we do not include species that do not appear in the two specified phylogenies.) The results are shown in Fig. 4. The least-squares regression equation set through the origin is:

Contrast in $\log_{10}(N) = 1.850$ contrast in $\log_{10}(C_R)$

($r^2 = 0.163$, $t_{38} = 2.758$, $p = 0.004$). Since the slope may be biased by the lowest outlier in the lower left quadrant of Fig. 4, we recalculated the regression without this contrast. The resulting regression equation through the origin is:

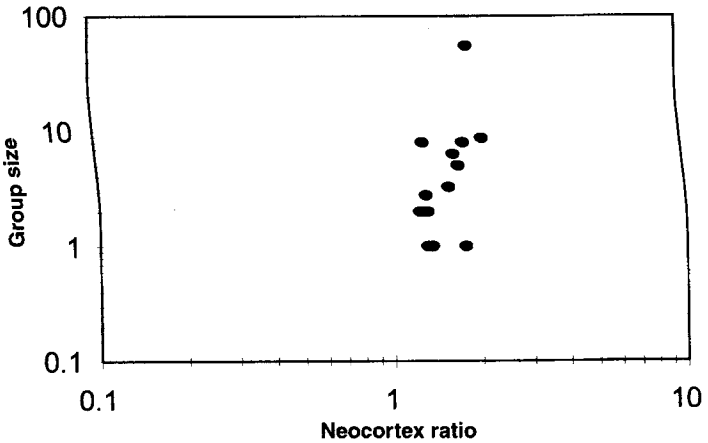


Fig. 3: Mean group size plotted against neocortex ratio estimated from total brain volume for canid, hyaenid and felid genera (see Table 2)

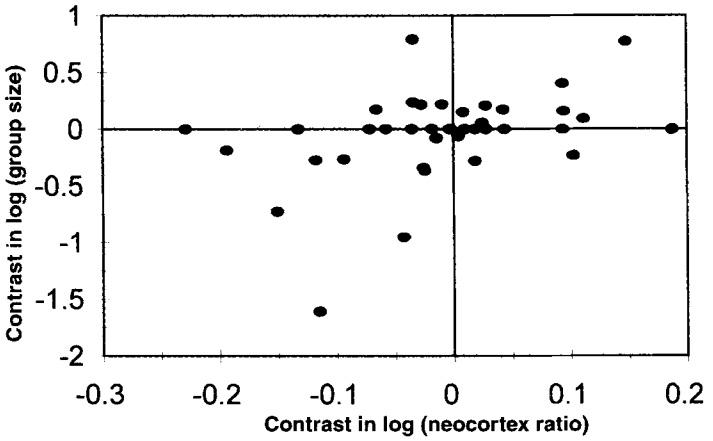


Fig. 4: Independent contrasts in group size plotted against contrasts in neocortex ratio (estimated from total brain volume) for 23 carnivore species (based on data given in Table 2). The species are: *Alopex* sp., *Canis mesomelas*, *Conepatus chinga*, *Conepatus humboldti*, *Crocuta crocuta*, *Eira* sp., *Galictis cuja*, *Galictis vittatus*, *Hyaena brunnea*, *Hyaena hyaena*, *Lutra canadensis*, *Lutra lutra*, *Lycaon* sp., *Martes martes*, *Martes pennanti*, *Meles meles*, *Mustela erminea*, *Mustela vison*, *Panthera leo*, *Procyon* sp., *Proteles* sp. and *Pteronura* sp. Phylogenies based on Bryant et al. (1993) and Zyll de Yong (1987) for mustelids and Wayne et al. (1989) for other species

$$\text{Contrast in } \log_{10}(N) = 1.292 \text{ contrast in } \log_{10}(C_R)$$

($r^2 = 0.129$, $t_{37} = 2.371$, $p = 0.013$).

The data for the insectivores are given in Table 3. The best-fit RMA regression equation set to individual generic means for the full dataset in Table 3 is:

$$\log_{10}(N) = 1.0763 \log_{10}(C_R) + 0.8498$$

($r^2 = 0.295$, $t_{33} = 2.715$, $p < 0.025$). A significantly improved fit is obtained by using absolute neocortex volume rather than the neocortex ratio:

$$\log_{10}(N) = 0.2782 \log_{10}(N_V) - 0.4778$$

($r^2 = 0.533$, $t_{33} = 3.840$, $p < 0.001$), where N_V is the absolute neocortex volume (in mm^3).

Fig. 5 plots mean group size against neocortex ratio for those insectivore genera that Stephan (1972) defines as 'basal' or 'advanced'. Inspection of the data in Fig. 5 suggests that it is primarily the 'advanced' insectivores that are responsible for generating the observed relationship between group size and neocortex size. Analysing the two groups separately confirms this: for the 'basal' insectivores, Spearman $r_s = -0.31$ ($n = 8$, $p = 0.25$) whereas for the 'advanced' insectivores $r_s = 0.78$ ($n = 9$, $p = 0.007$). The RMA regression equation for the 'advanced' insectivore genera only is:

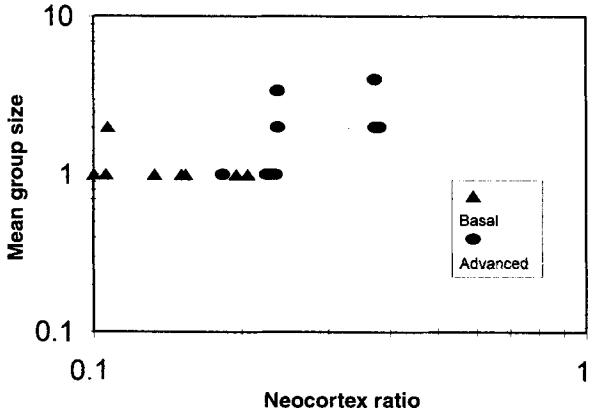


Fig. 5: Mean group size plotted against neocortex ratio for insectivore genera. Only those genera classified as 'basal' and 'advanced' insectivores by Stephan (1972) are shown (see Table 3)

$$\log_{10}(N) = 1.125 \log_{10}(C_R) + 0.899$$

Fig. 6 plots the generic means for carnivores and insectivores onto the distribution for primates (from Dunbar 1992; Fig. 1). Overall, the carnivore data map very closely onto the primate data (although, as noted earlier, the ursids and pandas tend to be marginal to the other groups). The RMA slope for the carnivores (Fig. 3: $b = 4.70$) is not significantly different from the slope for primates ($b = 3.389$) ($t_{25} = -1.95$, $p > 0.05$ two-tailed). The data for the insectivores,

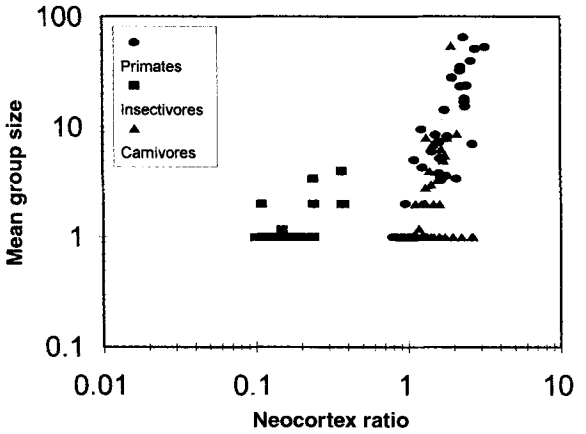


Fig. 6: Generic means for carnivores and insectivores plotted on the same graph as the equivalent data for primates (from Dunbar 1992). Data for carnivores from Table 2 and data for insectivores from Table 3

however, are displaced markedly to the left, having a much larger (positive) intercept than either the primates or the carnivores.

Discussion

The consistency of the relationships between group size and neocortex ratio in the various carnivore and insectivore samples suggests that the relationship first documented in primates by Dunbar (1992) may be characteristic of other advanced mammals. In addition, Barton (1993) and Tschudin (1996) presented evidence suggestive of a similar relationship in the bats and odontocete cetaceans, respectively.

The fact that the data for the various carnivore groups map quite neatly onto the end of the primate distribution strongly suggests that these two orders have brains that are structured along similar cognitive lines at the social level. In effect, the only reason primates live in larger groups than carnivores is that they have larger neocortices. Of particular interest, in this respect, is the spotted hyaena (*Crocuta crocuta*). This species is conspicuous among the carnivores both by virtue of its unusually large groups and its large neocortex size. It is significant that it is also characterized by unusually intense social behaviour involving ritualized greetings and pre-hunting 'ceremonies' (Kruuk 1972). In addition, the spotted hyaena is one of the few non-primate species to show evidence of coalition formation similar to that seen in primates (Zabel et al. 1992).

The insectivores are in striking contrast to these two orders in apparently lying far outside the distribution of both groups. There appears to be an important difference between primitive ('basal') and 'advanced' insectivores, in that the relationship between group size and neocortex size appears to be characteristic only of the 'advanced' genera. However, another explanation for the scatter in the insectivore data is that group sizes are especially likely to be poorly known in this order. As with the nocturnal semi-solitary prosimians (Dunbar 1992), it is conventionally assumed that group size is one for any species that appears to be solitary. However, in many cases, individual animals may have a detailed knowledge of a small subset of individuals within their immediate ranging area and may have very clearly defined relationships with these individuals. In prosimians, for example, these relationships are sometimes reflected in nest-sharing (Roberts 1971; Clark 1985; Bearder 1987; Kappeler & Ganzhorn 1993). It is possible that at least some of the larger brained insectivores live in larger groups (communities) than their solitary nature implies. The same explanation may also apply to the apparently solitary ursids and pandas. One implication of these findings is that more detailed studies of the social behaviour of these semi-solitary species might be profitable.

If it is accepted that insectivores are, in general, representative of the ancestral mammals, Fig. 6 suggests that there has been a significant grade shift in cognitive capabilities at some point within mammalian evolution leading to a different kind of relationship between animals. In part, this may reflect simply the transition between solitariness and sociality. However, it is also clear that there have been a series of grade shifts beyond this basal grade-shift in which increasing quantities

of neocortex-based processing power was needed to maintain group cohesion. Since prosimians appear to lie to the left of both the anthropoid primates (Dunbar & Joffe, unpubl. data) and the carnivores, it appears likely that there has been parallel convergent evolution within several mammalian orders as these evolved during more recent geological time periods.

Acknowledgements

We are grateful to M. Röhrs for providing us with unpublished data and to L. Aiello, R. Barton, J. Gittleman and an anonymous referee for their help and comments.

Literature Cited

Aiello, L. C. & Dunbar, R. I. M. 1993: Neocortex size, group size and the evolution of language. Curr. Anthropol. **34,** 184—193.

Barton, R. A. 1993: Independent contrasts analysis of neocortical size and socioecology in primates. Behav. Brain Sci **16,** 694—695.

—— 1996: Neocortex size and behavioural ecology in primates. Proc. Roy. Soc. B **263,** 173—177.

—— & Dunbar, R. I. M. 1996: Evolution of the social brain. In: Machiavellian Intelligence, Vol. 2 (Byrne, R. & Whiten, A., eds), Cambridge Univ. Press, Cambridge. pp. 240—263.

Bearder, S. 1987: Lorises, bushbabies, and tarsiers: diverse societies in solitary foragers. In: Primate Societies (Smuts, B., Cheney, D., Seyfarth, R., Wrangham, R. & Struhsaker, T., eds), Chicago Univ. Press, Chicago. pp. 11—24.

Bryant, H. N., Russell, A. P. & Fitch, W. D. 1993: Phylogenetic relationships within the extant Mustelidae (Carnivora): appraisal of the cladistic status of the Simpsonian subfamilies. Zool. J. Linn. Soc. **108,** 301—334.

Byrne, R. W. & Whiten, A. (eds) 1988: Machiavellian Intelligence. Oxford Univ. Press, Oxford.

Clark, A. B. 1985: Sociality in a nocturnal 'solitary' prosimian: *Galago crassicaudatus*. Int. J. Primatol. **6,** 581—600.

Deacon, T. W. 1990: Fallacies of progression indices in theories of brain-size evolution. Int. J. Primatol. **11,** 193—236.

Dunbar, R. I. M. 1992: Neocortex size as a constraint on group size in primates. J. human. Evol. **20,** 469—493.

—— 1993: Coevolution of neocortical size, group size and language in humans. Behav. Brain Sci. **16,** 681—735.

—— 1995: Neocortex size and group size in primates: a test of the hypothesis. J. Human Evol. **28,** 287—296.

—— 1998: The social brain hypothesis. Evol. Anthropol., in press.

Frahm, H., Stephan, H. & Stephan, M. 1982: Comparison of brain structure and volumes in Insectivora and Primates. I. Neocortex. J. Hirnforsch. **23,** 375—389.

Gittleman, J. L. 1986: Carnivore brain size, behavioural ecology and phylogeny. J. Mammal. **67,** 23—36.

—— 1989a: Carnivore group living: comparative trends. In: Carnivore Behavioural Ecology and Evolution (Gittleman, J. L., ed.)., Chapman & Hall, London. pp. 183—207.

—— (ed.) 1989b: Carnivore Behavioural Ecology and Evolution. Chapman & Hall, London.

Harcourt, A. H. & de Waal, F. (eds) 1992: Coalitions and Alliances in Humans and Other Animals. Oxford Univ. Press, Oxford.

Harvey, P. H. & Pagel, M. D. 1991: The Comparative Method in Evolutionary Biology. Oxford Univ. Press, Oxford.

Jerison, H. J. 1973: Evolution of the Brain and Intelligence. Academic Press, New York.

Joffe, T. & Dunbar, R. I. M. 1997: Visual and socio-cognitive information processing in primate brain evolution. Proc. Roy. Soc. Lond. B **264,** 1303—1307.

—— & —— 1998: How to measure brains. Evol. Anthropol., in press.

Kamiya, T. & Pirlot, P. 1988a: The brain of the lesser panda *Ailurus fulgens*: a quantitative approach. Z. Zool. Syst. Evol.-Forschg. **26,** 65—72.

—— & —— 1988b: The brain of the Malayan bear (*Helarctos malayanus*). Z. Zool. Syst. Evol.-Forschg. **26,** 225—235.

Kappeler, P. M. & Ganzhorn, J. U. (eds) 1993: Lemur Social Systems and their Ecological Basis. Plenum Press, New York.

Kruuk, H. 1972: The Spotted Hyaena. Univ. of Chicago Press, Chicago.

Marino, L. 1996: What can dolphins tell us about primate evolution? Evol. Anthropol. **5,** 81- 86.

Rayner, J. M. V. 1985: Linear relations in biomechanics: the statistics of scaling functions. J. Zool. Lond. **206,** 415—439.

Roberts, P. 1971: Social interactions in *Galago crassicaudatus*. Folia Primatol. **14,** 171- 181.

Röhrs, M. 1986a: Cephalisation, Telencephalisation und Neocorticalisation bei Mustelidae. Z. Zoo. Syst. Evol.-Forschg. **24,** 157—166.

—— 1986b: Cephalisation bei Caniden. Z. Zool. Syst. Evol.-Forschg. **24,** 300—307.

——, Ebinger, P. & Weidemann, J. 1989: Cephalisation bei Viverridae, Hyaenidae, Procyonidae und Ursidae. Z. Zool. Syst. Evol.-Forschg. **27,** 169—180.

Sawaguchi, T. & Kudo, H. 1990: Neocortical development and social structure in primates. Primates **31,** 131—145.

Stephan, H. 1972: Evolution of primate brains: a comparative anatomical investigation. In: Functional and Evolutionary Biology of Primates (Tuttle, R., ed.)., Aldine-Atherton, Chicago. pp. 155—174.

——, Baron, G. & Fons, R. 1984: Brains of Soricidae. II. Volume comparison of brain components. Z. Zool. Syst. Evol.-Forschg. **22,** 328—342.

——, —— & Frahm, H. D. 1991: Insectivora, with a Stereotaxic Atlas of the Hedgehog Brain. Springer-Verlag, Berlin.

——, Frahm, H. & Baron, G. 1981: New and revised data on volumes of brain structures in insectivores and primates. Folia Primatol. **35,** 1—29.

Tschudin, A. 1996: The use of neuroimaging in the assessment of brain size and structure in odontocetes. MSc thesis, Univ. of Natal, Durban.

Wayne, R. F., Benveniste, R. E., Janczewski, D. N. & O'Brien, S. J. 1989: Molecular and biochemical evolution of the Carnivora. In: Carnivore Behavioural Ecology and Evolution (Gittleman, J. L., ed.)., Chapman & Hall, London. pp. 465—494.

Willner, L. A. 1989: Sexual dimorphism in primates. PhD thesis, Univ. of London.

—— & Martin, R. D. 1985: Some basic principles of mammalian sexual dimorphism. In: Human Sexual Dimorphism (Ghesquiere, J., Martin, R. D. & Newcombe, F., eds). Taylor & Francis, London. pp. 1—19.

Wozencraft, W. C. 1989: The phylogeny of recent Carnivora. In: Carnivore Behavioural Ecology and Evolution (Gittleman, J. L., ed.)., Chapman & Hall, London. pp. 495—535.

Zabel, C. J., Glickman, S. E., Frank, L. G., Woodmansee, K. B. & Keppel, G. 1992: Coalition formation in a colony of prepubertal spotted hyaenas. In: Coalitions and Alliances in Humans and Other Animals (Harcourt, A. H. & de Waal, F., eds), Oxford Univ. Press, Oxford. pp. 113—136.

Zyll de Jong, C. G. 1987: A phylogenetic study of the Lutrinae (Carnivora; Mustelidae) using morphological data. Can. J. Zool. **65,** 2536—2544.

Received: June 24, 1996

Accepted: February 20, 1998 (W. Pflumm)